Feature drift report

Is tonight's open book still the population the uplift model learned from?

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TL;DR

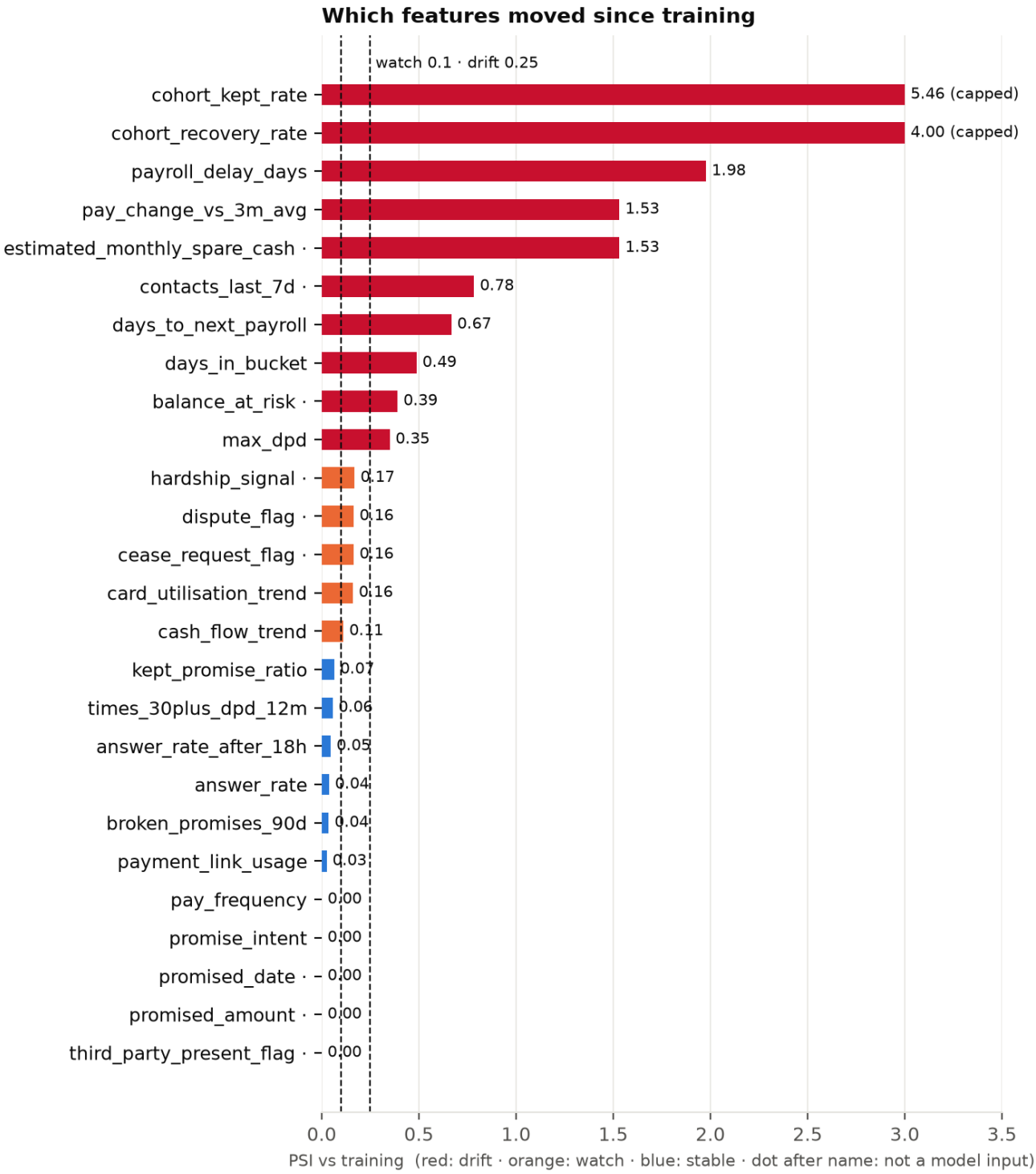
10 drift · 5 watch · 11 stable across the 26 registered features, comparing 690,062 training rows with the 96,000 cases served tonight (features as of 2026-10-03 06:00:00+00:00).

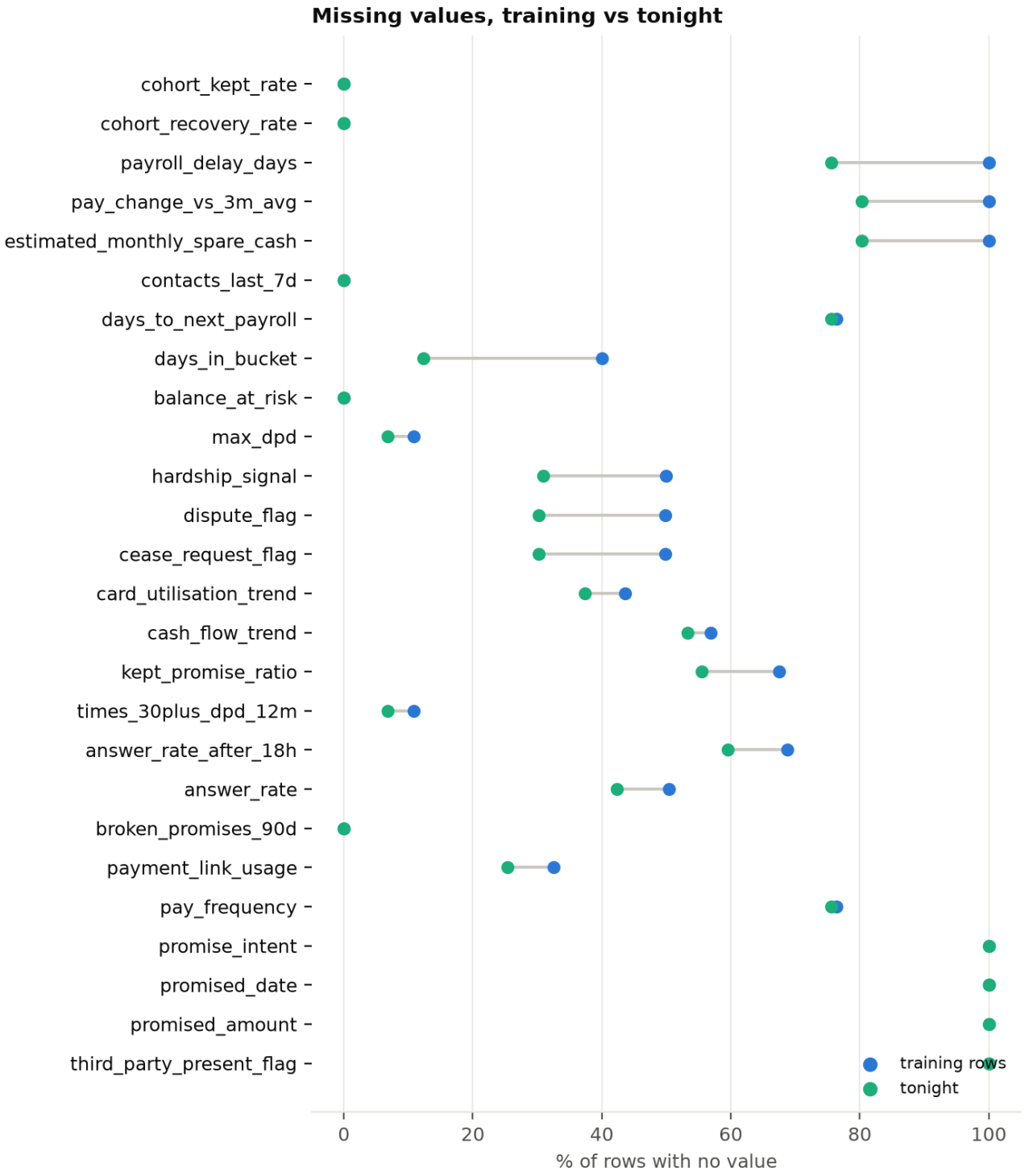
Two things are going on, and only one of them is a problem:

1. **The baseline is different by design.** Training rows are *every past decision since 2016*; tonight is *today's open book*. Open cases skew later-stage and bigger-balance, so delinquency features (days past due, balance at risk, days in bucket) and cohort rates look "drifted" just because it's a different slice of customers, not because customers changed.
2. **The real finding:** `payroll_delay_days`, `pay_change_vs_3m_avg`, `estimated_monthly_spare_cash` are **missing for every training row** but filled for ~20–25% of tonight's cases. The payroll history only lines up with recent decisions, so the model never got to learn from them. They're the heart of Kept (payday, spare cash), so the next retrain should backfill them for past decisions. Until then the rules and Kept sizing use them; the model just doesn't.

How to read PSI

PSI (population stability index) compares two distributions bin by bin: Σ (live% − train%) × ln(live% / train%). Numbers are cut at the training deciles, categories by value, and **missing values get their own bin**, so a feature that suddenly goes blank shows up too. Usual bank bands: **< 0.1 stable**, **0.1–0.25 watch**, **≥ 0.25 drift** (investigate, then retrain).


Which features moved since training
watch 0.1 · drift 0.25
cohort_kept_rate
cohort_recovery_rate
payroll_delay_days
pay_change_vs_3m_avg
estimated_monthly_spare_cash ·
contacts_last_7d ·
days_to_next_payroll
days_in_bucket
balance_at_risk ·
max_dpd
hardship_signal ·
dispute_flag ·
cease_request_flag ·
card_utilisation_trend
cash_flow_trend
kept_promise_ratio
times_30plus_dpd_12m
answer_rate_after_18h
answer_rate
broken_promises_90d
payment_link_usage
pay_frequency
promise_intent
promised_date ·
promised_amount ·
third_party_present_flag ·
5.46 (capped)
4.00 (capped)
1.98
1.53
1.53
0.78
0.67
0.49
0.39
0.35
0.17
0.16
0.16
0.16
0.11
0.07
0.06
0.05
0.04
0.04
0.03
0.00
0.00
0.00
0.00
0.00
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
PSI vs training (red: drift · orange: watch · blue: stable · dot after name: not a model input)



Every feature

Feature	PSI	Status	Missing (train)	Missing (tonight)	Model input
cohort_kept_rate	5.460	drift	0%	0%	yes
cohort_recovery_rate	4.003	drift	0%	0%	yes

Feature	PSI	Status	Missing (train)	Missing (tonight)	Model input
payroll_delay_days	1.977	drift	100%	76%	yes
pay_change_vs_3m_avg	1.532	drift	100%	80%	yes
estimated_monthly_spare_cash	1.532	drift	100%	80%	no
contacts_last_7d	0.784	drift	0%	0%	no
days_to_next_payroll	0.667	drift	76%	76%	yes
days_in_bucket	0.488	drift	40%	12%	yes
balance_at_risk	0.391	drift	0%	0%	no
max_dpd	0.350	drift	11%	7%	yes
hardship_signal	0.170	watch	50%	31%	no
dispute_flag	0.165	watch	50%	30%	no
cease_request_flag	0.165	watch	50%	30%	no
card_utilisation_trend	0.161	watch	44%	37%	yes
cash_flow_trend	0.112	watch	57%	53%	yes
kept_promise_ratio	0.065	stable	67%	55%	yes
times_30plus_dpd_12m	0.056	stable	11%	7%	yes
answer_rate_after_18h	0.046	stable	69%	60%	yes
answer_rate	0.038	stable	50%	42%	yes
broken_promises_90d	0.036	stable	0%	0%	yes
payment_link_usage	0.026	stable	32%	25%	yes
pay_frequency	0.000	stable	76%	76%	yes
promise_intent	0.000	stable	100%	100%	yes
promised_date	0.000	stable	100%	100%	no
promised_amount	0.000	stable	100%	100%	no
third_party_present_flag	0.000	stable	100%	100%	no

What we do with this

- **Nightly:** `kept drift` runs after `kept features`; a drifted *model input* that isn't explained by the open-book baseline is a reason to hold the new scores and retrain.
- **Next retrain:** backfill the three payroll features for past decisions, then re-check that their PSI and missing share line up.
- **Not a fairness check.** Drift says the population moved; the fairness report says whether groups get treated differently. Different questions.

Rebuild: `uv run python scripts/drift_report.py` (with `kept serve` stopped).